



Fig. 4: Circuit for interfacing with the sound card

0.6 ms. Then the 7-bit command is transmitted, followed by the 5-bit address.

Decoding

Decoding an IR signal sent by a transmitter means identifying the command and address information sent. Once you get that information, you can use the corresponding key in your projects. So what you would need is the demodulated signal, which is nothing but the pulse train shown in Fig. 2. From that, it is easy to read the required information.

Demodulation. Demodulation is nothing but removing the carrier and retrieving the pulse train. This can be easily done by direct use of photo modules like TSOP17xx from Vishay Telefunken.

'xx' in TSOP17xx represents the centre receiving frequency of the receiver. So for a 40kHz signal, you have to use TSOP1740 or TSOP1738 (for 38kHz), but gain may be slightly less.

The frequency response of the photo module is shown in Fig. 3.

Now what you get at the output is the pulse train but inverted.

Display. Next, give this pulse to the microphone input of the sound card of a PC. The circuit for interfacing with the sound card is shown in Fig. 4.

The circuit used is very simple. A 5V regulator IC is used to provide power to IR receiver TSOP1740 (TSOP1738 can also be used). Connect the output pin to the sound card input using a stereo pin.

R1 and R2 act as potential dividers and R2 can be varied to set the output voltage. Initially, keep it at the lowermost point and then increase the output level to a small safe value (around 0.25V peak would be fine).

How to use

Follow the steps below:

1. Configure the sound card so that there is a microphone input.
2. Connect the output of the circuit to the microphone input.

3. Switch on the power supply.

4. Press any button on the remote control so you can hear a tone through the speaker connected to the computer.

5. Start the Sound Recorder application as follows: Start→Programs→Accessories→Entertainment→Sound Recorder

6. Press the record button on the application and then press any key on the remote control you wish to decode.

7. Press the stop button on the Sound Recorder application.

8. Save the recorded sound by File→Save As, giving an appropriate name and path.

9. Now open the Nero WaveEditor application.

10. Open the previous saved file by pressing CTRL+O on the keyboard.

11. You can see the amplitude spectrum of the signal.

12. Zoom into the spectrum and view the pulse train.

Note that the output of the IR receiver is inverted, so what you see is also the inverted pulse train. Also, you won't be able to get the exact time span of high and low states, but if you are well aware of the protocol used, then it is very easy to identify the bit sequence. See the screenshots of the example.

Example

1. Open the Sound Recorder application



2. Record the signal by pressing any key on the remote control

